SHIPPING & SALES PERFORMANCE ANALYSIS

Project Overview

This project analyzes shipping and sales data to uncover insights into shipping performance, delivery time, sales trends, regional performance, and product categories. The goal is to identify shipping patterns, evaluate operational efficiency, and support data-driven business decisions using Excel, Python (Pandas), PostgreSQL, and Power BI.

Dataset Summary

- Rows: 9,800

- Columns: 18

- Dataset Type: Retail sales and shipping data

- Key Features:

  - Order Details: Row ID, Order ID, Order Date, and Ship Date.
  - Shipping Information: Ship Mode, shipping duration, and Region.
  - Customer Details: Customer ID, Customer Name, and Segment.
  - Geographical Details: Country, City, State, and Postal Code.
  - Product Details: Product ID, Category, Sub-Category, and Product Name.
  - Sales Information: Sales amount for each product/order record.

- Missing Data: 11 missing values in the Postal Code column.

- Duplicate Records: 0 duplicate rows found.

- Data Transformation: Shipping duration can be calculated using the difference between Ship Date and Order Date.

Exploratory Data Analysis using Python

We began with data preparation and cleaning in python:

- **Data Loading** Imported the dataset using in pandas.
- **Initial Exploration:** Used df.info( ) to check structure and .describe( ) for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 19 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9800 non-null   int64
 1   Order ID       9800 non-null   object
 2   Order Date     9800 non-null   object
 3   Ship Date      9800 non-null   object
 4   Total Time     9800 non-null   int64
 5   Ship Mode      9800 non-null   object
 6   Customer ID    9800 non-null   object
 7   Customer Name  9800 non-null   object
 8   Segment        9800 non-null   object
 9   Country        9800 non-null   object
 10  City           9800 non-null   object
 11  State          9800 non-null   object
 12  Postal Code    9800 non-null   int64
 13  Region         9800 non-null   object
 14  Product ID     9800 non-null   object
 15  Category       9800 non-null   object
 16  Sub-Category   9800 non-null   object
 17  Product Name   9800 non-null   object
 18  Sales          9800 non-null   float64
dtypes: float64(1), int64(3), object(15)
memory usage: 1.4+ MB
```

	Row ID	Total Time	Postal Code	Sales
count	9800.000000	9800.000000	9800.000000	9800.000000
mean	4900.500000	3.961122	55217.343265	230.769079
std	2829.160653	1.749614	32066.750532	626.651906
min	1.000000	0.000000	1040.000000	0.440000
25%	2450.750000	3.000000	23223.000000	17.250000
50%	4900.500000	4.000000	57551.000000	54.490000
75%	7350.250000	5.000000	90008.000000	210.605000
max	9800.000000	7.000000	99301.000000	22638.480000

- **Missing Data Handling: Identified missing values in the Postal Code column and investigated the corresponding city and state information.**
- **Column Standardization: Renamed column headers to snake_case for improved readability and consistency.**
- **Feature Engineering: Created a shipping_days column by calculating the difference between the Ship Date and Order Date.**
- **Date Transformation: Converted Order Date and Ship Date columns into datetime format for time-based analysis.**
- **Data Quality Checks: Checked for duplicate records, missing values, and data type inconsistencies to improve data quality.**
- **Exploratory Data Analysis: Analyzed sales trends, product categories, customer segments, regional performance, and shipping modes using Python and Pandas.**
- **Database Integration: Loaded the cleaned DataFrame into PostgreSQL for SQL-based analysis.**
- **Data Visualization: Used Power BI to develop interactive dashboards for analyzing sales performance and shipping efficiency.**

Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Customer Segment

Compare total sales generated by Consumer, Corporate, and Home Office Customers.

	Segment text	total_revenue numeric
1	Consumer	1148060.51
2	Corporate	688494.14
3	Home Offi...	424982.32

2. Revenue by Shipping Mode

Compare sales associated with each shipping method.

	Ship Mode text	total_revenue numeric
1	Standard Cla...	1340831.64
2	Second Class	449914.04
3	First Class	345572.26
4	Same Day	125219.03

3. Revenue by Region

Identify which geographical region contributes the most sales.

	Region text	total_revenue numeric
1	West	710219.77
2	East	669518.85
3	Central	492646.90
4	South	389151.45

4. Revenue by Product Category

Compare sales from Furniture, Office Supplies, and Technology.

	Category text	total_revenue numeric
1	Technology	827455.94
2	Furniture	728658.75
3	Office Suppli...	705422.28

5. Monthly Revenue Trend

Identify the top months with the highest and lowest sales.

	month_start date	monthly_revenue numeric
1	2015-01-01	14205.71
2	2015-02-01	4519.92
3	2015-03-01	55205.83
4	2015-04-01	27906.86
5	2015-05-01	23644.30

6. Average Shipping Time by Shipping Mode

Determine which shipping method has the longest average delivery duration

	Ship Mode text	avg_shipping_days numeric
1	Standard Cla...	5.01
2	Second Class	3.25
3	First Class	2.18
4	Same Day	0.04

7. Top 10 States by Revenue

Identify the states contributing the most sales.

	State text	total_revenue numeric
1	California	446306.49
2	New York	306361.07
3	Texas	168572.47
4	Washington	135206.87
5	Pennsylvan...	116276.76
6	Florida	88436.55
7	Illinois	79236.57
8	Michigan	76136.07
9	Ohio	75130.43
10	Virginia	70636.72

8. Top 10 Customers by Revenue

Find your highest-value customers based on total sales.

	Customer ID text	Customer Name text	total_revenue numeric
1	SM-20320	Sean Miller	25043.07
2	TC-20980	Tamara Chand	19052.22
3	RB-19360	Raymond Buch	15117.35
4	TA-21385	Tom Ashbrook	14595.62
5	AB-10105	Adrian Barton	14473.57
6	KL-16645	Ken Lonsdale	14175.23
7	SC-20095	Sanjit Chand	14142.34
8	HL-15040	Hunter Lopez	12873.30
9	SE-20110	Sanjit Engle	12209.44
10	CC-12370	Christopher Cona...	12129.08

9. Revenue by Sub-Category

Discover which product sub-categories contribute the most and least sales.

	Sub-Category text	total_revenue numeric
1	Phones	327782.49
2	Chairs	322822.75
3	Storage	219343.37
4	Tables	202810.77
5	Binders	200028.82
6	Machines	189238.68
7	Accessories	164186.70
8	Copiers	146248.07
9	Bookcases	113813.25
10	Appliances	104618.38
11	Furnishings	89211.98
12	Paper	76828.34
13	Supplies	46420.29
14	Art	26705.42
15	Envelopes	16128.02
16	Labels	12347.71
17	Fasteners	3001.93

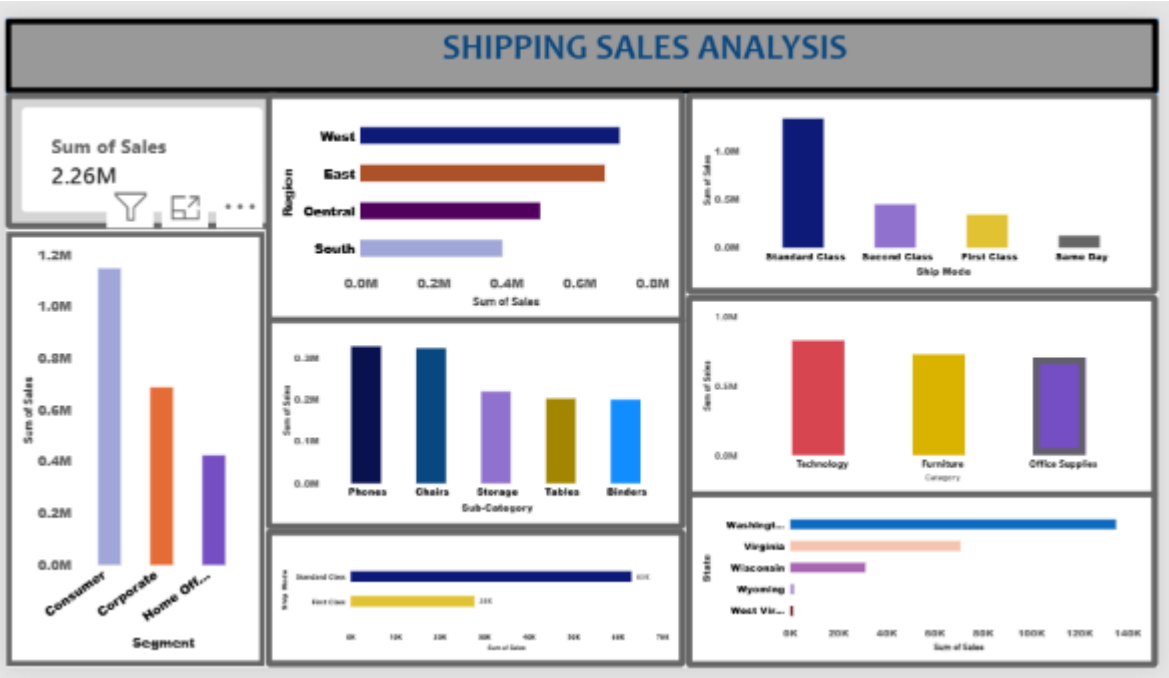
10. Average Order Value by Customer Segment

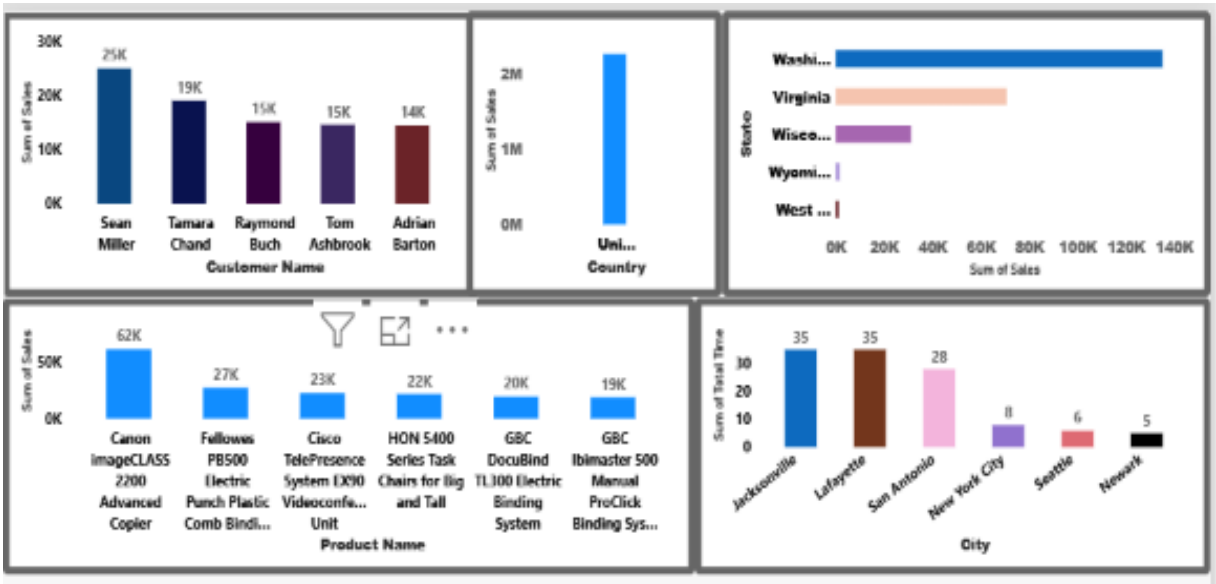
Compare average revenue per distinct order across customer segments.

	Segment text	avg_order_value numeric
1	Home Offi...	475.37
2	Corporate	461.77
3	Consumer	452.53

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.





6. Business Recommendations

- **Optimize Shipping Modes** – Promote cost-effective shipping options while maintaining reliable delivery times.
- **Improve Delivery Efficiency** – Identify delays and reduce shipping time to improve operational performance.
- **Focus on High-Revenue Regions** – Strengthen marketing and distribution strategies in regions generating the highest sales.
- **Promote Best-Selling Products** – Highlight top-performing product categories and sub-categories to increase revenue.
- **Strengthen Customer Retention** – Reward repeat customers through loyalty programs and personalized offers.
- **Optimize Inventory Planning** – Use monthly sales trends to forecast demand, manage stock levels, and prepare for peak sales periods.